DSA 8420 Spring 2025
Homework 8

Due March 25, 2025

1. Write down the dual problem of the following linear programs. (You can use Table 1 at the end of the homework for help.)

   (a) (10 points)

$$
\begin{array}{rl}
\max & 5x_1 + 6x_2 + 4x_3 \\
s.t. & -x_1 + 2x_2 \leq 1 \\
& 3x_1 - x_2 \geq 2 \\
& 3x_2 + x_3 = 3 \\
& x_2 \geq 0,\ x_3 \leq 0
\end{array}
$$

   (b) (10 points)

$$
\begin{array}{rl}
\min & x_1 + 2x_2 - 3x_3 \\
s.t. & -3x_1 + x_2 + 2x_3 = 16 \\
& 2x_1 + 4x_2 + 3x_3 \geq 20 \\
& x_2 + 2x_3 \leq 6 \\
& x_1 \geq 0,\ x_2 \leq 0
\end{array}
$$

2. Consider the following linear program.

$$
\begin{array}{rl}
\max & 3x_1 + 10x_2 + 5x_3 + 11x_4 + 6x_5 + 14x_6 \\
s.t. & x_1 + 7x_2 + 3x_3 + 4x_4 + 2x_5 + 5x_6 = 42 \\
& x_1,\ \ldots,\ x_6 \geq 0
\end{array}
$$

(a) (4 points) Write down the dual problem.

(b) (4 points) Solve the dual problem by observation/simple logic.

(c) (4 points) With your solution to (b), what is the optimal value of the primal problem? Why?

3. Consider the following primal-dual pair.

$$\begin{array}{lrl} \max & x_1 + 2x_2 + 5x_3 + x_4 & \\ s.t. & x_1 + 2x_2 + \ \ x_3 - x_4 \leq 20 & \\ & -x_1 + \ \ x_2 + \ \ x_3 + x_4 \leq 12 & \\ & 2x_1 + \ \ x_2 + \ \ x_3 - x_4 \leq 30 & \\ & x_1,\ x_2,\ x_3,\ x_4 \geq 0 & \end{array} \qquad \text{(P)}$$

$$\begin{array}{lrl} \min & 20\pi_1 + 12\pi_2 + 30\pi_3 & \\ s.t. & \pi_1 - \ \ \pi_2 + \ 2\pi_3 \geq 1 & \\ & 2\pi_1 + \ \ \pi_2 + \ \ \pi_3 \geq 2 & \\ & \pi_1 + \ \ \pi_2 + \ \ \pi_3 \geq 5 & \\ & -\pi_1 + \ \ \pi_2 - \ \ \pi_3 \geq 1 & \\ & \pi_1,\ \pi_2,\ \pi_3 \geq 0 & \end{array} \qquad \text{(D)}$$

(a) (4 points) Show that $\bar{\mathbf{x}} = (10, 0, 16, 6)$ is feasible to (P). What is the objective value of (P) at $\bar{\mathbf{x}}$?

(b) (4 points) Show that $\bar{\boldsymbol{\pi}} = (0, 3, 2)$ is feasible to (D). What is the objective value of (D) at $\bar{\boldsymbol{\pi}}$?

(c) (4 points) Use the above results to explain why $\bar{\mathbf{x}}$ is optimal to (P) and $\bar{\boldsymbol{\pi}}$ is optimal to (D).

(d) (6 points) Verify that the complementary slackness conditions are satisfied by $(\bar{\mathbf{x}}, \bar{\boldsymbol{\pi}})$.

Maximization Problem		Minimization Problem
Constraints		Variables
$\leq$	$\leftrightarrow$	≥ 0
$\geq$	$\leftrightarrow$	≤ 0
$=$	$\leftrightarrow$	*unrestricted*
Variables		Constraints
≥ 0	$\leftrightarrow$	$\geq$
≤ 0	$\leftrightarrow$	$\leq$
unrestricted	$\leftrightarrow$	$=$

Table 1: Primal-dual relationships